

Bálits Péter Márton

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EDUCATION

Eötvös Loránd University

Bachelor of Arts in Computer Science

Budapest, HU

Sep. 2021 – July 2026

- Thesis: wrote a 40K lines-of-code etcd clone in Go from scratch, using HashiCorp's raft library and Kubernetes

Czech Technical University

Erasmus semester at the Faculty of Informatics

Prague, CZ

Sep. 2024 – Febr. 2025

- Built a CSV spreadsheet engine in Scala focusing on expression lexing, parsing and cell arithmetics

EXPERIENCE

Backend Developer

Fornax ICT

Oct. 2025 – July. 2026

Budapest, HU

- Developed and maintained backend services using Java, Spring Boot and PostgreSQL.
- Contributed to internal projects with mature codebases, designed APIs, optimized query performances.

PROJECTS

Kave | Distrubted key-value store with MVCC, Leases, Watches and Oblivious Transfer

Jan. 2026 – May 2026

Go, Svelte

- * Distributed Raft cluster over Kubernetes with fault tolerance, rolling updates, pod distribution and prometheus metrics.
- * MVCC as the fundamental concurrency primitive, used for global monotonic revisions, snapshot isolation and transactions.
- * Leases for attaching TTLs to keys, with a background expiration checker.
- * Watches for streaming real time changes to subscribed keys over WebSocket.
- * HTTP API endpoints, with follower-to-leader proxying.
- * docker-compose for local development, stress and smoke tests, GitHub Actions for CI/CD.

Logregator | Time-series log aggregator over TCP

May 2026 – Aug. 2026

Rust, Tokio, Glommio

- * Async log aggregator designed append-heavy workloads over TCP.
- * Custom time-series LSM tree storage engine, with SSTables for persistence and WAL for crash recovery.
- * Label indexed logs inspired by Grafana Loki for fast queries.
- * Custom benchmarking suite for fine-grain performance tuning and visualizing.
- * Used cargo workspaces with subcrates for storage engine, tcp server/client, internal benchmarking.

TECHNICAL SKILLS

Languages: Rust, Go, Scala, Java, Python, C++, SQL, TypeScript

Technologies: Linux, git, make, cargo, Docker, Kubernetes, GitHub Actions

Systems, protocols: Raft consensus, TCP/IP, HTTP, distributed storage, write-ahead logs

Spoken languages: Hungarian (native), English (fluent), German (B2)